

4 (a) Vaccination programmes are widely used to help control the spread of infectious diseases.

(i) State why some diseases are described as infectious.

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..... [1]

(ii) A person was given an injection to give protection against infectious disease **E**. The person had **not** previously been infected with disease **E**. 26 days later, a second injection to give protection against disease **E** was given.

The concentration in the blood of the antibody specific to disease **E** was measured over a period of 60 days from the time of the first injection.

Fig. 4.1 shows the concentration of the antibody in the blood over the period of 60 days.

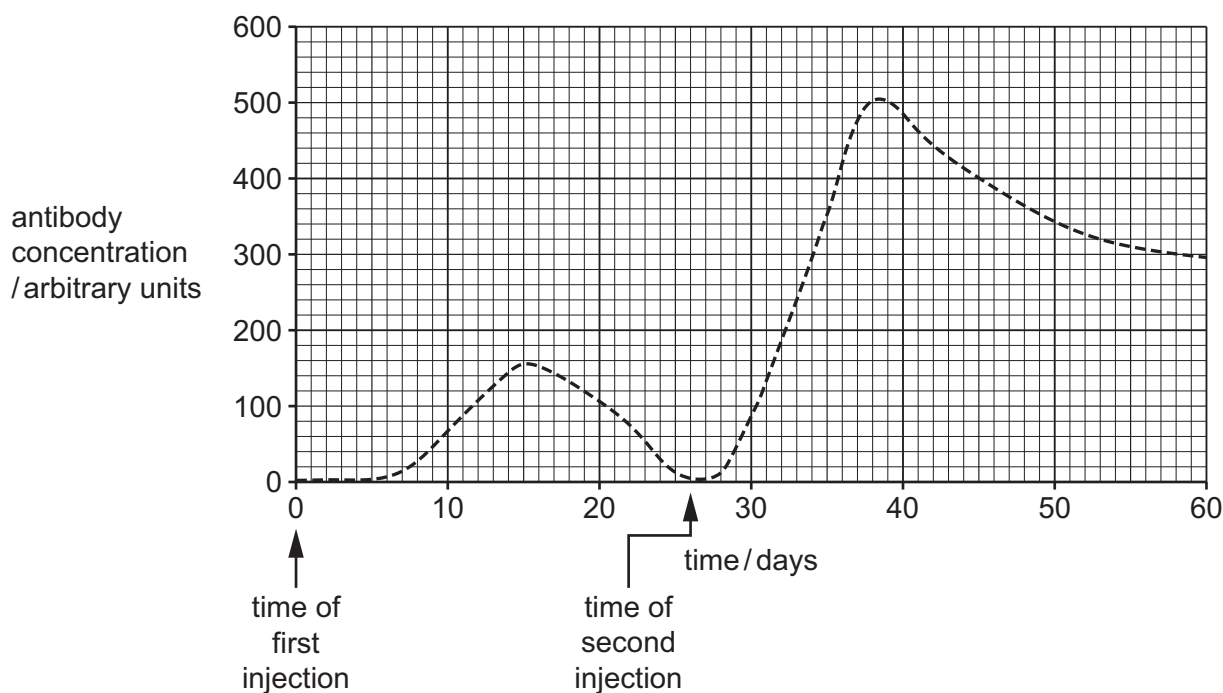


Fig. 4.1



The concentration of the antibody in the blood after the second injection was higher than the concentration after the first injection.

Explain why the concentration of the antibody in the blood was higher after the second injection than after the first injection.

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- (b) A second person was given a different injection to give protection against another infectious disease, infectious disease **F**. The person had **not** previously been infected with disease **F**.

The concentration in the blood of the antibody specific to disease **F** was measured over a period of 60 days from the time of the injection.

Fig. 4.2 shows the concentration of the antibody in the blood over the period of 60 days.

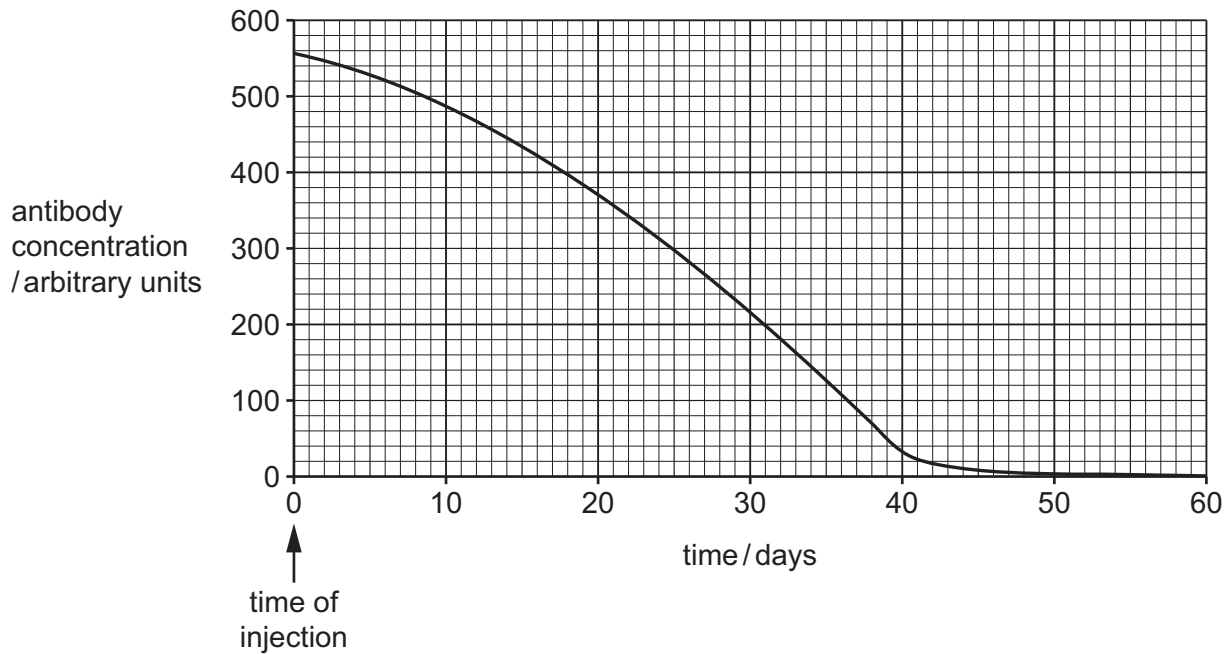


Fig. 4.2

- (i) State the type of immunity that results from the injection given as protection against disease **F**.

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- (ii) Describe features of the type of immunity resulting from the injection given as protection against disease **F**.

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..... [2]



Question 4 continues on page 18.

(c) In 2021, the number of new cases of tuberculosis (TB) was estimated to be 10.6 million.

(i) Name the bacterium that causes TB.

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(ii) TB is often treated with several different drugs at the same time. This is necessary to kill multiple drug resistant (MDR) strains of bacteria. This treatment is usually lengthy, taking 6 months or more to make sure all bacteria are killed.

Researchers investigated how the bacteria that cause TB react to the presence of rifampicin. Rifampicin is an antibiotic often used in the successful treatment of TB.

- Researchers combined rifampicin with a coloured marker dye and added the coloured rifampicin to a culture of living bacteria.
- When initially viewed under the microscope, the researchers could see that the coloured rifampicin was present inside the cytoplasm of the bacterial cells.
- Hours later, the coloured rifampicin was **not** visible inside the bacterial cells but was visible in the medium surrounding the bacterial cells.
- The researchers concluded that rifampicin was being pumped out of the bacterial cells through the cell surface membrane.
- Further research showed that when some drugs commonly used to treat indigestion were added to bacterial cultures containing coloured rifampicin, the coloured rifampicin stayed inside the bacterial cytoplasm.

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

Suggest **and** explain how knowledge gained from this research could improve TB treatment **and** reduce the chance of rifampicin-resistant bacteria developing.

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[Total: 12]